

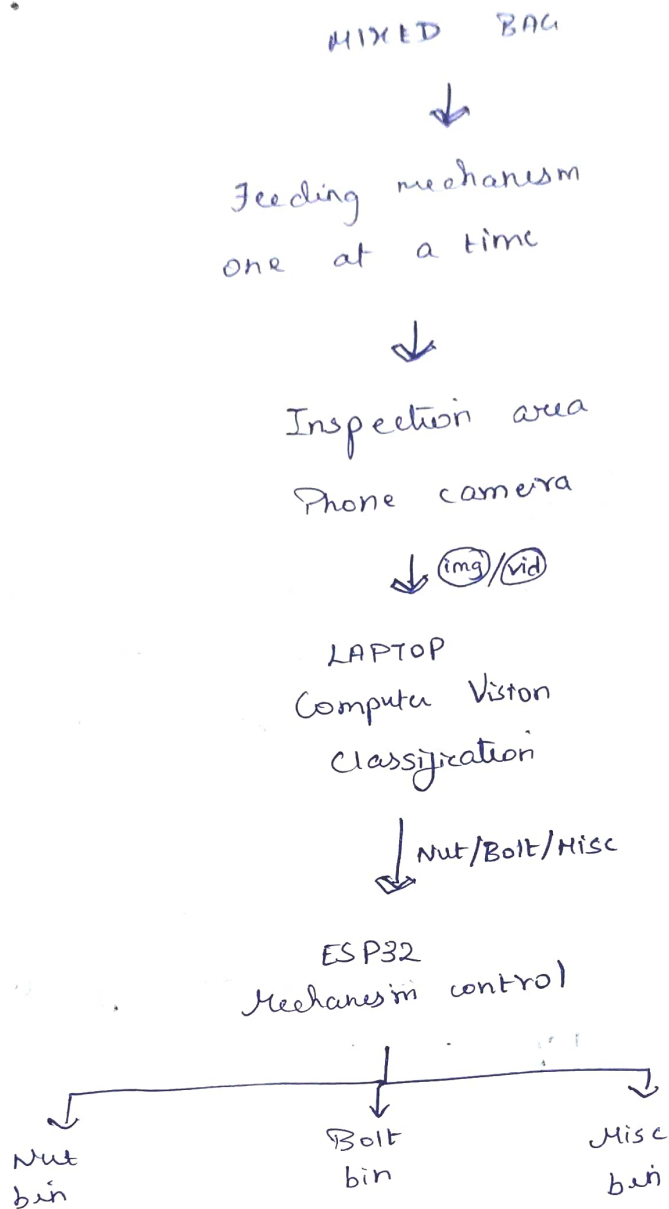
INTELLIGENT

SORTING

SYSTEM

Mechanical feeding → Phone camera → Classification → ESP32 sorting mech.

Overall build:



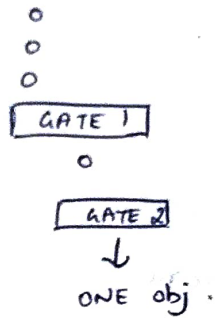
challenge: Exactly take one object at a time from mixed bag & bring it to inspection position

Possible mechanisms to tackle this

- Vibratory feeder
- Rotating disc
- Escapement mechanism
- Inclined belt.

[Choose one]

1. Vibratory feeder - obj sits on vibrating surface causes obj to move along a controlled vibration path until obj are separated.
2. Rotating disc - obj are placed on rotating disc
3. Escapement mech. - gates operate alternately so that one obj released at a time.

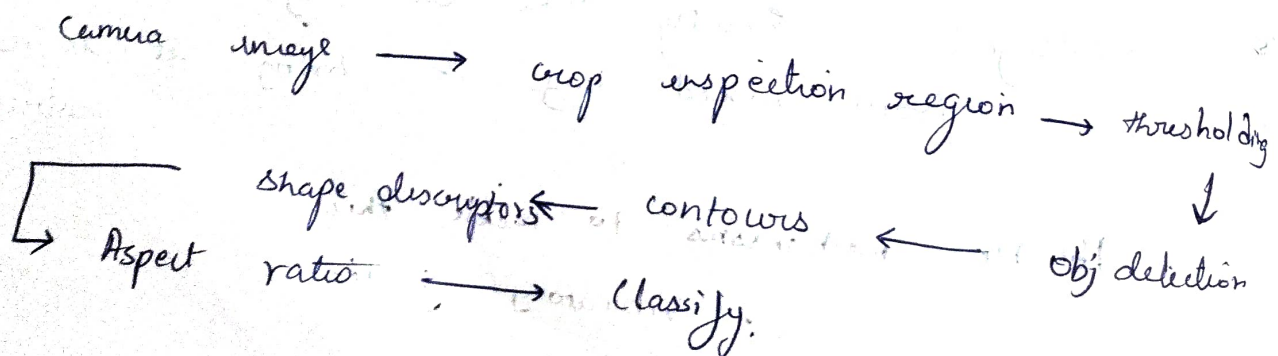


- Inclined belt - obj placed on moving belt, some geometry would be needed at the end to ensure only one obj. enters inspection area.

camera must be phone - don't use ESP32 camera
 ESP32 only controls motors, feeders, communicate with laptop, etc.

Computer Vision System :

Option A : Classic CV



Option B: trained model

Images → dataset → training → model → classify

Say classifier says nut 51%
bolt 49%

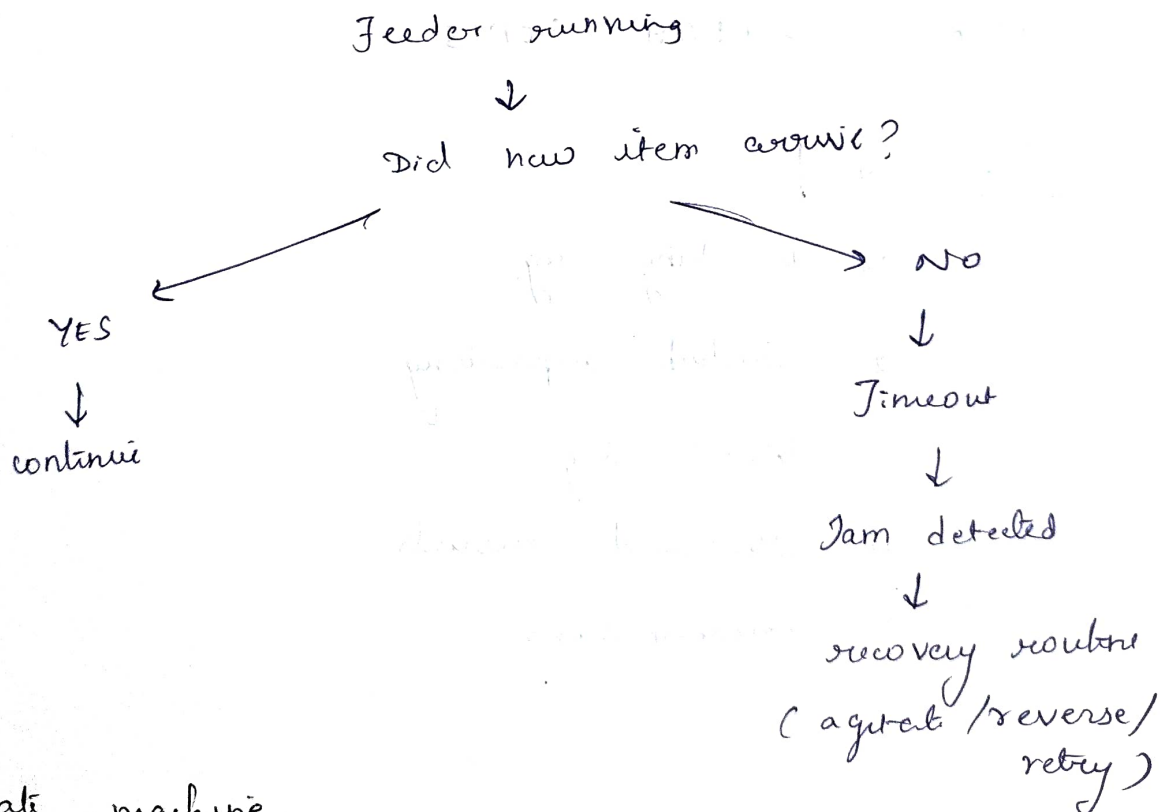
so classify as nut
(confidence > threshold)

find threshold experimentally

After identification, send to respective bin

- servo operated flap
- servo gate
- sliding mech.
- Rotating diverts
- Multiple gates

* JAM detection



* State machine

Dashboard must show idle, feeding, imaging, sorting, jammed, etc.

Dashboard must show counts & stats

* Analyse the performance of sys by running tests & finding confusion matrix

* Perform tests in diff. lightings

eg. test duration = 5 mins

obj - sorted = 62

Throughput = 12.4 items/min

Jams = 4

auto recoveries = 3

Manual intervention = 1

* ESP-IDF is the firmware to be used

Check: STRETCH GOALS

SUBMISSION

1. Marking sig.
2. Github repository
3. Build log
4. Measured results
5. Presentations